Computer Organization Homework 1 (Due on 2023/10/5)

1. Assume a 200mm diameter wafer has a cost of 12, contains 84 dies, and has 0.045
defects/cm2. Assume a 140mm diameter wafer has a cost of 15, contains 90 dies, and has 0.010 defects/cm2.
(a) Find the yield for both wafers (20%)
(b) For the 200mm diameter wafer, if the number of dies per wafer increased by 15% and the defect per area unit increases by 15%. Find the die area and yield. (10%)

2. Describe the steps that transform a program written in a high-level language such as C into a representation that is directly executed by a computer processor. (10%)

3. Consider two different implementations of the same instruction set architecture.
The instructions can be divided into five classes according to their CPI (class A, B, C, D), P1 with a clock rate of 2.5GHz and CPIs of 1,2,3,3 and P2 with a clock rate of 3GHz and CPI of 2,2,2,2. Given a program with a dynamic instruction count of 1.0e8 instructions divided into classes as follows: 10% class A, 20% class B, 50% class C, 20% class D
(a) What is the ratio of CPI for P1/P2? (10%)
(b) The result of the benchmark running on the machine has an instruction count of 2.5E12, and execution time of 900s, and a reference time of 9000s. Find the CPI if the clock cycle time is 0.25ns.(10%)
(c) Find the increase in CPU time if number of instructions of the benchmark is increased by 20% and the CPI is increased by 5%(15%)

4. Assume for arithmetic, load/store and branch instructions, a processor has CPIs of 3, 12, and 5, respectively. Also assume that on a single processor a program requires the execution of 2.56x109 arithmetic instructions, 1.28x109 load/store instructions, and 256 million branch instructions.

(a) By how much must we improve the CPI of load/store instructions if we want the program to run two times faster? (10%)

(b) By how much is the execution time of the program improved if the CPI of arithmetic and branch instructions is reduced by 50% and the CPI of load/store is reduced by 20%? (15%)